

★ WEEK 16 · PART 2 of 3

The Fair Comparison: Out-of-Sample Testing

One Asset, Three Families, One Honest Grade

Hold-out testing · walk-forward · the metrics a desk actually uses

Choosing a family is only half the work; proving it is the rest. This part builds the capstone comparison: fit all three families on the past, grade them on the future they never saw, and score them the way a trading desk would.

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PART 2 · THE FAIR COMPARISON

Fit on the Past, Grade on the Future

The capstone pipeline, hold-out and walk-forward testing, the look-ahead bias that fakes great results, the metrics that matter in finance, and how to read the scoreboard without fooling yourself.

From Choosing to Proving

Part 1 chose; Part 2 tests

1

We have three families

Classical, state-space, and Bayesian - each with a forecast shape suited to a different decision.

2

But a choice needs evidence

The competitions taught us not to trust taste or in-sample fit. We must grade the families on data they never saw.

3

So we build a fair comparison

Apply all three to one asset, hold out a test period, and score them identically against a naive benchmark.

Part 2 in one line: a fair, out-of-sample comparison - the same asset, the same test, the same metrics.

Apply All Three to One Asset

The shape of the project, end to end

The capstone takes one asset and runs it through all three families. Each is fit on a training period and asked to forecast a held-out test period. Then a single, honest scoreboard compares them - and ties back to a trading decision.

What goes in

- One asset's return series.
- A clean split into train and test.
- One representative model per family.

What comes out

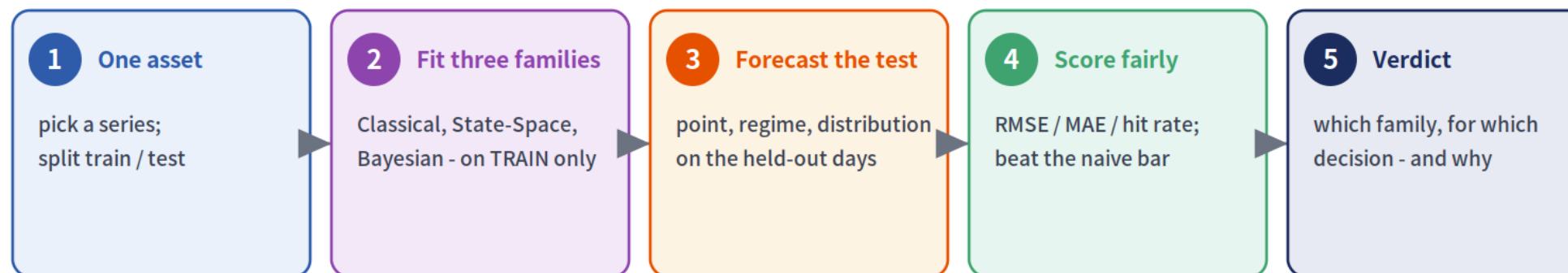
- A forecast from each family on the test.
- A scoreboard on identical metrics.
- A decision-driven verdict.

The Capstone as a Pipeline

One asset in; three families fit on the training data only; each forecasts the held-out test; all are scored on the same metrics against the naive bar; and the verdict follows the decision. Two rules keep it honest

The Capstone as a Pipeline

One asset in, three fitted families, one held-out grade, one honest verdict out.



The two rules that keep it honest

1. Fit on the past only - never let a model touch the test data before it is graded (no look-ahead bias).
2. Compare every family on the SAME held-out days, against the naive benchmark, with the same metrics.

The verdict is decision-driven: the winner is the family whose output your decision actually needs.

The Fair Comparison: Hold-Out Testing

Fit on the past, grade on the future it never saw

The heart of an honest comparison is the hold-out. Reserve the most recent slice of data as a test set, fit every model on the earlier part only, then grade each on the reserved slice. No model ever sees its own exam.

How it works

- Split the series into train and test.
- Fit each model on the training part.
- Forecast, then compare to the held-out truth.

Why it's fair

- Every model faces the same unseen test.
- In-sample fit can't flatter anyone here.
- The score reflects real forecasting skill.

Fit on the Past, Grade on the Future

The rule behind every honest test

Training

- Estimate all parameters on the past only.
- Let each model do its best on what it can see.
- This is where fitting happens - and stops.

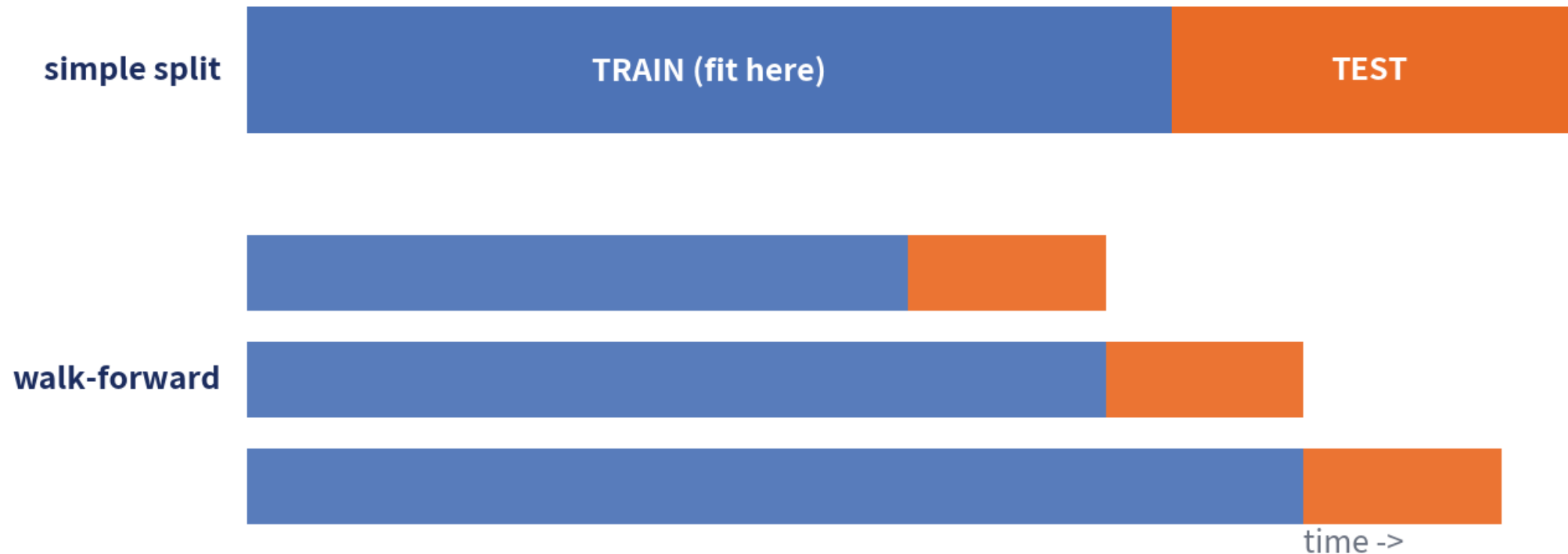
Testing

- Freeze the models; forecast the test period.
- Compare forecasts to the realized values.
- The grade is on data used for nothing else.

Train, Test, and Walk Forward

The simple version reserves one test slice at the end. Walk-forward does it repeatedly - fit, test one step, roll the window forward - so the grade averages over many honest out-of-sample forecasts rather than one lucky split

Fit on the past, grade on the future it never saw - once, or rolling forward



Walk-Forward Validation

One split can be lucky; many cannot

The idea

- Fit, forecast the next step, then roll forward.
- Re-fit as the window advances through time.
- Collect many out-of-sample forecasts.

Why it's better

- It averages over many test points, not one.
- It mimics how you'd actually trade in real time.
- A single split can flatter or punish by luck.

Look-Ahead Bias

The one mistake that fakes great results

Look-ahead bias is letting a model use information it could not have had at the time - a future value, a full-sample mean, a scaler fit on the test set. It produces gorgeous backtests that collapse in live trading. It is the single most common way to fool yourself.

How it sneaks in

- Scaling with statistics from the whole sample.
- Using a value that wasn't known yet.
- Tuning choices on the test set.

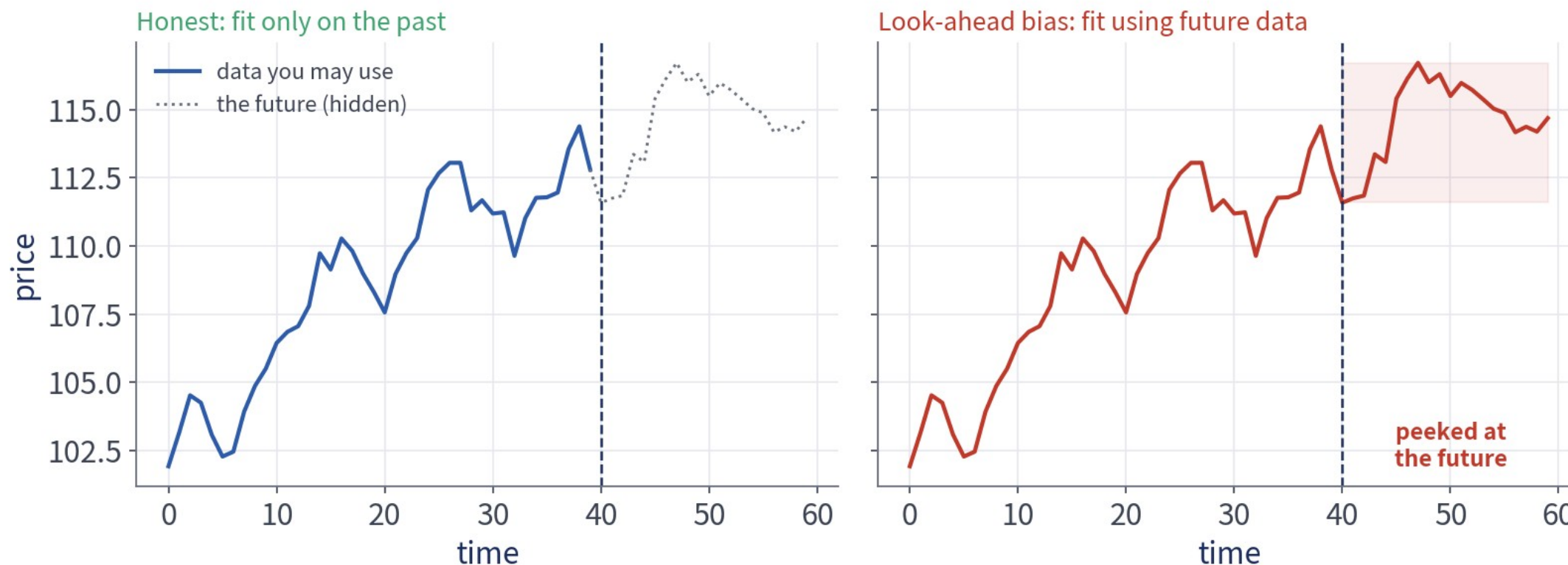
What it costs

- Backtests that look far too good.
- Results that vanish out of sample.
- Confidence in a strategy that doesn't work.

The Bias That Fakes Great Results

On the left, the honest setup uses only the past to fit. On the right, the model peeks at the future - and any statistic computed over the whole series does the same. Never let a model touch the data it will be graded on

The one bias that fakes great results: never let the model see its own test



How to Avoid Look-Ahead

A few habits that save you

Do this

- Compute every statistic on the training data only.
- Fit scalers and transforms on train, apply to test.
- Timestamp everything - use only what was known.

Never this

- Never normalize with full-sample mean or variance.
- Never tune hyperparameters on the test set.
- Never let a future bar leak into a past feature.

Metrics That Matter in Finance

Score forecasts the way a desk actually would

Not all errors are equal. A desk cares about the size of the error, whether the direction was right, and whether an honest uncertainty was reported. We use a small set of complementary metrics - and read them together, not one alone.

Three angles

- Size: how far off was the point forecast?
- Direction: did you get the sign right?
- Honesty: was the stated uncertainty calibrated?

Read them together

- A model can win on size but lose on direction.
- A tight point can hide an overconfident band.
- No single number tells the whole story.

Accuracy: RMSE and MAE

How far the point forecast lands

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_t (r_t - \hat{r}_t)^2}, \quad \text{MAE} = \frac{1}{n} \sum_t |r_t - \hat{r}_t|$$

What they measure

- RMSE: penalizes big misses heavily.
- MAE: the plain average miss.
- Both: lower is better, on the test set.

How to use them

- Report both next to the naive benchmark.
- RMSE if large errors hurt most.
- MAE if every miss counts the same.

Direction and P&L

Getting the sign right, and the money

$$\text{hit rate} = \frac{1}{n} \sum_t \mathbf{1}[\text{sign}(\hat{r}_t) = \text{sign}(r_t)], \quad \text{Sharpe of the signal's P\&L}$$

Why direction matters

- For a trading signal, the sign drives the P&L.
- A model can have low error yet wrong direction.
- The hit rate scores the sign directly.

The bottom line

- Ultimately, grade the signal's simulated P&L.
- A good forecast should make money out of sample.
- Report the strategy's Sharpe, not just the error.

Honesty: the Log Score

Reward a well-calibrated distribution

$$\text{log score} = \frac{1}{n} \sum_t \log p(r_t \mid \text{model}) \quad \text{rewards honest uncertainty, not just the point}$$

What it does

- It scores the whole predictive distribution.
- It rewards putting probability on what happens.
- It punishes overconfidence and vagueness alike.

Why it matters here

- RMSE ignores the band; the log score doesn't.
- It's how you grade the Bayesian family fairly.
- Calibrated uncertainty finally gets credit.

Which Metric for Which Decision

Score what the decision cares about

If you will...

- Book a number: use RMSE or MAE.
- Trade a direction: use hit rate and P&L.
- Size risk: use the log score / calibration.

The rule

- Grade on the quantity you'll actually act on.
- A model can win one metric and lose another.
- Choose the metric before you see the scores.

Reading the Forecast Scoreboard

Lower error wins - but mind the fine print

With the metrics in hand, the scoreboard looks simple: lowest error wins. But read the fine print. Are the gaps large enough to matter? Did every model beat naive? Is the winner winning on the metric your decision cares about?

What to check

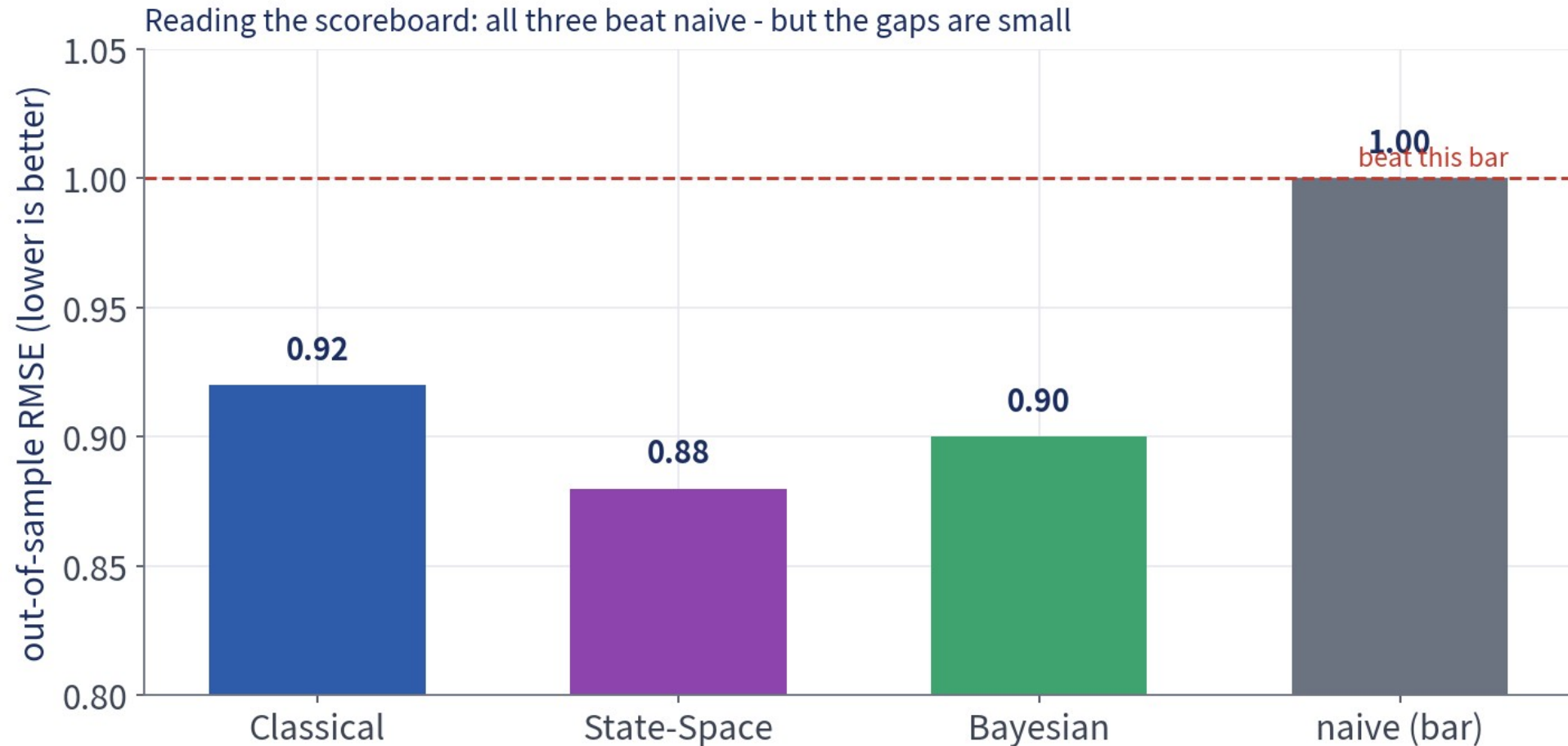
- Did each family clear the naive bar?
- Are the differences meaningful or noise?
- Which metric is the verdict resting on?

What to resist

- Don't crown a winner on a razor-thin gap.
- Don't ignore a family that fits your decision.
- Don't let one lucky split decide it.

Reading the Scoreboard

All three families beat the naive benchmark here, but the gaps between them are small. That's the honest, common outcome - and it means the tie is broken by which forecast shape the decision needs, not by a hair of RMSE



What We're Really Grading Is a Trade

Every metric maps to a trading action

The scoreboard is not an academic exercise - each metric decides part of a trade. Accuracy sets how much to trust the point that fixes the hedge; the hit rate is the P&L of the direction; the log score grades the band that sizes the risk. We compare families to pick a trade, not a number

Metric -> trading job

- RMSE / MAE: trust in the point that sets the hedge.
- Hit rate: the direction that drives the P&L.
- Log score: the band that sizes the position.

So the winner is

- The family whose output the trade actually uses.
- Not the lowest decimal, but the best decision.
- Part 3 turns this straight into a pairs trade.

Small Gaps, Big Caveats

Don't over-read a thin margin

Why gaps are often small

- Financial returns are close to unpredictable.
- Much of the signal is common to all models.
- A tiny RMSE edge can be pure noise.

So be humble

- Check whether the gap survives another split.
- Prefer the simpler model when it's a near tie.
- Let the decision, not decimals, break the tie.

The Honest-Comparison Checklist

Six boxes to tick before you trust a result

Setup

- One asset, one clean train/test split.
- All models fit on train only.
- No statistic computed on the test.

Scoring

- Same metrics for every family.
- Every model compared to naive.
- The verdict tied to the decision.

Build the Comparison Harness

Direct Claude Code to grade the three families

GOAL Turn Part 1's three fitted families into a fair, out-of-sample scoreboard.

[INPUT] Prompt · Claude Code

Write and run `compare_families.py` that:

1. loads one asset and makes a train/test split (and, optionally, a walk-forward loop);
2. fits Classical, State-Space, and Bayesian models on TRAIN only - no statistic from the test;
3. forecasts the test period with each;
4. scores RMSE, MAE, hit rate, and a naive benchmark - plus a log score for the Bayesian;
5. prints a scoreboard and flags any tie.

Guard against look-ahead: fit scalars on train only.

How to work with it

- Same split for every model.
- Beat the naive bar first.
- Read several metrics, not one.
- Break ties by the decision.

Deliverable: `compare_families.py` + a scoreboard on identical metrics, guarded against look-ahead bias.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What makes a comparison fair?

**Same held-out test,
same metrics, no
leakage**

Q2 What is look-ahead bias?

**Using data the model
couldn't have had**

Q3 Why read several metrics together?

**A model can win one
and lose another**

All three come from the hold-out method, the look-ahead trap, and the finance metrics.

Three Pitfalls to Avoid

Where fair comparisons go wrong

Common mistakes

- "A great backtest means a great strategy."
- "The lowest RMSE always wins."
- "One train/test split is enough."

The reality

- Look-ahead bias fakes great backtests - guard against it.
- Read direction and calibration too, and mind small gaps.
- Walk forward so one lucky split can't decide it.

Three Things to Carry Forward

Part 2 in one screen

1

Fit on the past, grade on the future

Reserve a test period, fit every model on the earlier data only, and score each on the slice it never saw. Walk-forward averages over many honest forecasts instead of one lucky split.

2

Look-ahead bias is the cardinal sin

Any statistic or value the model couldn't have had at the time fakes great results that vanish live. Compute everything on train only, and never tune on the test.

3

Score like a desk, and read the fine print

Use size, direction, and calibration together, always against the naive bar. Gaps are often small - so let the decision, not a hair of RMSE, break the tie.

Next — Part 3: the final project end to end, from comparison to a trading decision, and the course close.